Week 6 · Mon

HTML Forms & Validation

The gateway for every user interaction

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Why Do Forms Matter?



Login

Every "Sign In" button submits a form to verify your identity.



Search

Google, YouTube, every search bar is a single-input form.



Registration

Account creation, surveys, checkout — all rely on HTML forms.

Forms are the #1 way users send data to your server.

The <form> Element

```
<!-- Basic form skeleton -->
<form
  action="/register"
  method="POST"
  id="reg-form"
  autocomplete="on">

  <label for="name">Full Name</label>
  <input type="text" id="name" name="name">

  <label for="email">Email</label>
  <input type="email" id="email" name="email">

  <button type="submit">Register</button>
</form>
```

action

Where the data is sent

method

How it is sent (GET or POST)

id / name

Target via JS or labels

novalidate

Skip browser auto-check

autocomplete

Browser fill suggestions

GET vs POST — Which to Use?

GET

- ✅ Data visible in the URL
- ✅ Bookmarkable, shareable
- ✅ Cached by browsers
- ✅ Great for search forms
- ❌ Bad for passwords / private data
- ❌ URL length limited (~2000 chars)

POST

- ✅ Data hidden in request body
- ✅ Handles large payloads
- ✅ Supports file uploads
- ✅ Correct for login / signup
- ❌ Not cacheable
- ❌ Back-button re-submits

QUICK QUIZ

Which HTML element wraps all form controls?

A <form action="..." method="...">

B

C <section>

D <input>

💬 Discuss with your neighbor — then we'll reveal the answer!

✅ ANSWER

Which HTML element wraps all form controls?

A **<form action="..." method="...">**

B

C <section>

D <input>

<form> is the semantic container that holds all inputs and defines where and how data is sent via action and method attributes.

Input Types — HTML5 Gives You Many

`type="text"`

Single-line text

e.g. name, city

`type="email"`

Validates @ syntax

e.g. user@domain.com

`type="password"`

Masks characters

e.g. ••••••••

`type="number"`

Numeric spinner

e.g. age: 1–120

`type="date"`

Date picker UI

e.g. 2026-03-15

`type="range"`

Slider control

e.g. volume 0–100

`type="checkbox"`

Multi-select (boolean)

e.g. interests

`type="radio"`

Single-select group

e.g. gender

`type="select"`

Dropdown list (<select>)

e.g. country

QUICK QUIZ

Which input type shows a native date picker in the browser?

A type="calendar"

B type="date"

C type="datetime"

D type="picker"

Discuss with your neighbor — then we'll reveal the answer!

✅ ANSWER

Which input type shows a native date picker in the browser?

A type="calendar"

B type="date"

C type="datetime"

D type="picker"

type="date" opens the browser's built-in calendar UI with no JavaScript needed. No other option is a valid HTML5 input type.

HTML5 Validation Attributes

```
<!-- required: cannot submit empty -->
<input type="text" name="name" required>

<!-- pattern: 9-digit student ID -->
<input type="text" name="sid"
  pattern="\d{9}"
  title="9-digit student ID"
  required>

<!-- min/max: age 18-99 -->
<input type="number" name="age"
  min="18" max="99" required>

<!-- maxlength: bio limit -->
<textarea name="bio"
  maxlength="200"></textarea>
```

No JS required!

`required`

Field cannot be empty

`pattern`

RegEx must match

`min / max`

Number or date range

`maxlength`

Cap character count

`minlength`

Minimum character count

QUICK QUIZ

A form field has minlength='8'. What happens if the user types only 5 characters and clicks Submit?

A The form submits normally

B The browser blocks submission and shows a validation tooltip

C JavaScript throws an error

D The field turns red automatically

Discuss with your neighbor — then we'll reveal the answer!

✅ ANSWER

A form field has minlength='8'. What happens if the user types only 5 characters and clicks Submit?

A The form submits normally

B The browser blocks submission and shows a validation tooltip

C JavaScript throws an error

D The field turns red automatically

Browser-native validation blocks the submit and shows a tooltip — no JS needed if required and minlength are set. The browser handles this entirely.

The <label> Element

```
<!-- ✅ EXPLICIT linking (preferred) -->
<!-- for= must match id= exactly   -->
<label for="email">Email address</label>
<input type="email" id="email"
       name="email" required>

<!-- ✅ IMPLICIT wrapping -->
<label>
  Email address
  <input type="email" name="email">
</label>

<!-- ❌ NO label → screen readers -->
<!--    announce as "edit text"   -->
<input type="email" name="email"
       placeholder="Email">
```

Critical for accessibility

`for="id"`

Links label to input

`click area`

Click label to focus input

`screen readers`

Reads label text aloud

`implicit wrap`

Wrap input inside label

<fieldset> + <legend>

```
<!-- Grouping personal info -->
<fieldset>
  <legend>Personal Information</legend>

  <label for="fname">First Name</label>
  <input type="text" id="fname"
         name="fname" required>

  <label for="lname">Last Name</label>
  <input type="text" id="lname"
         name="lname" required>
</fieldset>

<fieldset>
  <legend>Contact Details</legend>
  <!-- email, phone here -->
</fieldset>
```

Group related fields

`<fieldset>`

Draws a box around a group

`<legend>`

Label for the group

`disabled`

Disables all fields inside

`form=`

Associate with external form

CSS Pseudo-classes for Forms

```
/* Green border when valid */
input:valid {
  border: 2px solid #27C93F;
}

/* Red border + icon when invalid */
input:invalid {
  border: 2px solid #FF5F56;
  background: url("x.svg")
    no-repeat right 0.5rem center;
}

/* Focus ring for keyboard nav */
input:focus {
  outline: 3px solid #F7DF1E;
  outline-offset: 2px;
}

/* Hide error until touched */
input:placeholder-shown { border: none; }
```

Style validity states

`:valid`

Input passes all constraints

`:invalid`

Input fails a constraint

`:focus`

Input is currently active

`:placeholder-shown`

Placeholder visible (empty)

`:required`

Has required attribute

QUICK QUIZ

Which CSS selector applies styles to an input that FAILED browser validation?

A input:error

B input:invalid

C input:wrong

D input:failed

Discuss with your neighbor — then we'll reveal the answer!

✅ ANSWER

Which CSS selector applies styles to an input that FAILED browser validation?

A input:error

B input:invalid

C input:wrong

D input:failed

:invalid is a CSS pseudo-class that matches inputs whose value violates their constraints (required, pattern, min, max, minlength, etc.).

LIVE DEMO

Build a student registration form together

Follow along in VS Code

Student Registration Form

```
<form action="/register" method="POST"
      id="reg-form" novalidate>
  <fieldset>
    <legend>Student Info</legend>
    <label for="sname">Full Name</label>
    <input type="text" id="sname"
           name="sname" required
           minlength="2" maxlength="50">

    <label for="sid">Student ID</label>
    <input type="text" id="sid"
           name="sid" required
           pattern="\d{9}"
           title="9-digit number">

    <label for="semail">Gachon Email</label>
    <input type="email" id="semail"
           name="semail" required>
  </fieldset>
  <button type="submit">Submit</button>
</form>
```

Follow along!

- Remove required and test
- Add pattern="\d{9}"
- Test with 8 digits (fail)
- Test with 9 digits (pass)
- Show browser tooltip

Form Accessibility Quick-Wins



Always use <label>

Link with for/id. Never rely on placeholder alone.



Keyboard navigation

Tab through all fields, Enter submits. Test it!



Error messages

Describe the problem: "Must be a valid email address".



Mobile input types

type="email" gives the @ keyboard on smartphones.

Summary & Homework

`<form>`	`Input types`	`Validation attrs`
action + method (GET/POST)	text, email, password, date…	required, pattern, min/max

`<label>`	`<fieldset>`	`CSS pseudo-classes`
for/id linking — always!	Group related controls	:valid, :invalid, :focus

HOMEWORK

Add a contact form to your personal site: name, email, subject, message (textarea), and a send button. Apply at least 3 HTML5 validation attributes and style :valid/:invalid states with CSS.

https://muxiddin19.github.io/WebPro/week7/WebPro_Week6_Mon_StudyGuide.html

https://muxiddin19.github.io/WebPro/week7/WebPro_Week6_Mon_VideoLecture.html

https://muxiddin19.github.io/WebPro/week7/WebPro_Week6_Mon_Forms_v2.pdf